Electronic Sources for Network Analysis

The discipline of social network analysis traditionally faced significant challenges in data collection, requiring considerable ingenuity and effort from researchers.

Traditional Data Collection Methods and Their Limitations

- **Observation:** This method involved close involvement of the researcher, which could influence the data collection process.
- **Standardized Surveys:** While minimizing observer influence, surveys rely on active engagement from participants. In contemporary times, achieving high response rates is problematic due to the proliferation of surveys. Even when surveys can be enforced, such as within companies, concerns arise about the spontaneity and genuineness of responses.
- **Repetitive Nature:** Both observation and surveys require multiple repetitions to effectively study network dynamics in detail.
- **Labor-Intensive and Costly:** These manual methods are extremely labor-intensive, often consuming up to 50% of a project's time and resources in network analysis. This immense effort often forces researchers to reanalyze existing datasets rather than collecting new ones, limiting contributions to the field.

Creative Solutions: Reusing Existing Electronic Records To overcome these limitations and find less costly empirical data, network analysts began to creatively reuse existing electronic records of social interaction that were not originally created for network analysis.

- **Examples of Reused Sources:**
  - **Scientific Communities:** Studied using publication or project databases that show collaborations among authors or institutions [BJN+02, GM02].
  - **Corporate Innovation:** Networks of innovation analyzed using official databases on corporate technology agreements [Lem03].
  - **News Archives:** A source for studies on diverse topics, from social-cognitive networks in politics [vAKOS06] to the structure of terror organizations [Kre02].
- **Benefits:** These sources often support dynamic studies through historical analysis.
- **Drawbacks:** Access to publication and patent databases, media archives, and legal/financial records often comes with a significant price tag.

The Internet: A Vast, Diverse, Dynamic, and Free Data Source The Internet stands out as a unique data source because it is vast, diverse, dynamic, and freely accessible to all.

- **e-Social Science:** A rapidly emerging field relies entirely on data collected from electronic networks and online information sources. This allows for the complete automation of the data collection process.
- **Scale of Studies:** Many e-social science studies are orders of magnitude larger than those based on observation or surveys, even without relying on commercial databases. They cover diverse social settings and exploit the dynamics of electronic data for longitudinal analysis.
- **Limitations:**
  - **Data Availability:** What is not on the Web cannot be extracted from it, meaning some social settings require offline study methods.
  - **Technological Accuracy:** There are technological limits to the accuracy of methods relying on Information Extraction. These methods need evaluation before use for network analysis.

Types of Electronic Sources for Network Analysis:

3.1 Electronic Discussion Networks These sources involve communication data, providing insights into social interactions.

- **Email Data:**
  - **Hewlett-Packard's Information Dynamics Labs:** Conducted a prominent series of studies analyzing communication among their employees using corporate email archives [TWH03]. They recreated discussion networks by linking individuals who exchanged a minimum number of emails, filtering out one-way relationships.
  - **Insights Gained:** Studying email networks helped identify leadership roles and both formal (organizationally dictated) and informal (cross-organizational) communities. These findings were verified through interviews with employees.
  - **Further Research:** This dataset was used to verify formal models of information flow based on epidemic models [WHAT04]. Another study revisited the local search problem, finding that additional knowledge about contacts (like physical location and organizational position) improved search efficiency compared to simply passing messages to the most connected neighbor.
  - **Limitations (Privacy):** Despite its versatility, email-based communication network studies are limited by privacy concerns.

    For example, in the HP case, researchers had to ignore message content, and the dataset could not be shared with the community.

- **Public Forums and Mailing Lists:**
    - **Overcoming Privacy Issues:** These sources can be analyzed without the same privacy concerns as personal emails.
    - **USENET Newsgroups:** Marc Smith and colleagues published extensively on visualizing and analyzing USENET newsgroups, which predate web-based forums [Smi99].
    - **W3C Mailing Lists:** Peter Gloor and colleagues analyzed archives of mailing lists from the World Wide Web Consortium (W3C), an organization known for its transparency [GLDZ03].
    - **Methodology:** Similar to email analysis, Gloor used message headers to automatically recreate discussion networks within working groups.
    - **Key Contribution:** Gloor developed a dynamic visualization of discussion networks that allowed quick identification of key discussion moments that activated the entire group, not just a few members. He also performed comparative studies across groups based on emerging structures over time.
    - **Future Potential:** Such studies could be extended by analyzing the role of networks in decision-making using electronic voting records. Applying emotion mining techniques from AI to message content could recover agreements and disagreements among committee members, a task almost impossible manually due to the sheer volume of emails (over ten thousand per working group).

3.2 Blogs and Online Communities These represent a dynamic and structured source of social interaction data.

- **Blogs as Communication Webs:** While often seen as "personal publishing," blogs are much more, enabling communication through comments and reactions to others' posts. These interactions foster dynamic communities, evidenced by syndicated blogs, blog rolls (lists of discussion partners), and even real-world meetings.
- **Features for Social Network Extraction:** Blogs offer features like comments, links from/to other blogs, quotes, and timestamps (crucial for studying network dynamics and information spread) for network extraction.
- **Structured Data (RSS Feeds):** Blogs are appealing research targets due to the availability of structured electronic data in RSS (Rich Site Summary) feeds. RSS feeds contain post text and valuable metadata like timestamps, enabling dynamic analysis.

- **Information Diffusion:** Researchers like Kumar et al. and Gruhl et al. have studied information diffusion in blogs using this data [KNRT03, GGLNT04]. Early work by Efimova and Anjewierden also studied blogs from a communication perspective [AE06]. Adar and Adamic visualized such communication [AA05b].
- **Turning Point (2004 US Election):** The 2004 US election campaign marked a significant shift in blog research, demonstrating blogs' capacity to build networks among activists and supporters [AG05]. This expanded blog analysis beyond marketing interests to the study of stable, long-distance social network creation and maintenance.
- **Online Community Spaces and Social Networking Services:**
    - **Direct Socialization:** Platforms like MySpace and LiveJournal directly facilitate socialization through features like friend lists, group joining, messaging, and photo sharing.
    - **Youth Culture Studies:** Their common usage by younger demographics provides excellent opportunities to study changes in youth culture.
    - **Examples:** Paolillo, Mercure, and Wright characterized the LiveJournal community based on exposed electronic data about user interests and social networks [PMW05]. Backstrom et al. studied LiveJournal data to understand the influence of structural properties on community formation and growth, and how changes in membership relate to discussion topics [BHKL06]. These studies exemplify how directly available electronic data enables longitudinal analysis of large communities (over 10,000 users) and go beyond structural properties to user interests.
    - **Data Access and Privacy:** While LiveJournal uniquely exposes data in a semantic format for research, most online social networking services (e.g., Friendster, Orkut, LinkedIn) closely guard their data, even from users, often through terms of use that limit user rights despite user ownership of the data.
    - **FOAF Network:** A technological alternative is the FOAF (Friend-of-a-Friend) network, where profiles are stored on user websites and linked via hyperlinks. However, FOAF currently lacks tools for profile creation/maintenance and useful services. Preliminary studies suggest FOAF exhibits similar characteristics to other online social networks [PW04, DZFJ05].

3.3 Web-Based Networks The vast content of Web pages offers an inexhaustible source for social network analysis.

- **Advantages:** It is vast, diverse, free to access, and often more up-to-date than specialized databases.

- **Disadvantages:** Information quality varies significantly, and reusing it for network analysis poses technical challenges. Efficient web mining typically requires commercial search engines, which often limit queries per day.
- **Features for Extracting Social Relations:**
  - **Links:** The linking structure of the Web serves as a proxy for real-world relationships, as authors choose to link to relevant and authoritative sources.
    - **Drawback:** Direct links between personal pages are very sparse due to the Web's increasing size and the dominance of searching over browsing. Individuals invest less effort in creating/updating personal links.
    - **Higher Aggregation Levels:** Most studies based on web linking structures analyze relationships at higher levels of aggregation, such as the web connectivity of entire scientific institutions for webometrics (web-based scientometrics). For example, the EICSTES project investigated this for scientific institutions. Extracting fine-grained networks from individual homepages is difficult due to link sparsity and the "homepage search problem" (locating an individual's homepage from their name/description).
  - **Co-occurrences:** The co-occurrence of names in web pages is a more frequent phenomenon and can be taken as evidence of relationships.
    - **Requirement:** Extracting relationships based on co-occurrence requires "web mining" (applying text mining to web pages), typically using statistical methods combined with content analysis.
    - **ReferralWeb Project (Kautz et al., mid-90s):**
- **Pioneer Work:** First tested web mining for social network extraction [KSS97].
- **Goal:** To build a tool for "referral chaining," finding experts close to a user through a chain of referrals (e.g., "experts on simulated annealing at most three steps away").
- **Application Domain:** Primarily scientific (finding peer-reviewers) and later corporate (recommending experts at AT&T).
- **Methodology:** The system was bootstrapped with names of famous AI researchers. It extracted connections via co-occurrence analysis using search engines (like Altavista) to collect page counts for individual names and co-occurrences. This involved shallow parsing, not counting indirect references (e.g., "the president of the United States" not associated with George Bush unless his name appeared).
- **Tie Strength Calculation (Jaccard-coefficient):** Calculated by dividing the number of co-occurrences by the number of pages for the two names individually. This is the ratio of the intersection of pages (where both

names appear) to their union [Sal89]. The value ranges from zero (no co-occurrences) to one (only co-occurrences, no separate mentions). A fixed threshold determined if a tie existed.

- **Support Filtering:** It is assumed Kautz also filtered ties based on "support" (absolute number of pages) because Jaccard-coefficient is a relative measure and can yield spurious results with very low absolute page counts (e.g., two names co-occurring once, yielding a coefficient of one, but too low to be evidence of a tie).
- **Expertise Extraction:** Capitalized phrases in search results that were not proper names were extracted as expertise.
- **Network Growth:** The system grew in two ways: through proper name extraction from web documents (similar to snowballing in network analysis) and by allowing users to register themselves.
- **Evaluation:** Kautz did not formally evaluate the extracted networks' accuracy against real-world networks. However, he noted the system performed well as a recommender, though recommendations could be "astonishingly accurate, or absolutely ridiculous". He suggested transparency (showing evidence and confidence levels) to maintain user trust. He also noted that personal pages could be more up-to-date than official corporate records for locating experts, and scientific settings often lack central expert systems.

Improvements and Variations on Kautz's Method (Author's Work) The author's work applied Kautz's basic method with some modifications:

- **Targeted Social Network Extraction:** Starting with a given list of names, the author's work consults the search engine for possible ties between *all pairs* of names.
- **Scalability Challenges:** This pairwise approach leads to a quadratic increase in the number of queries, which is costly and limited by search engine query caps. Solutions like Matsuo et al.'s method (extracting possible contacts from individual name searches) significantly reduce queries with minimal loss [MHT+06].
- **Different Co-occurrence Measures:**
    - **Jaccard-coefficient Disadvantage:** It unfairly penalizes ties between individuals whose names occur frequently on the Web and less popular individuals (e.g., a famous professor and their PhD student). The professor's name might appear on most pages mentioning the student, but not vice versa.
    - **Asymmetric Variant:** The author uses an asymmetric variant, dividing the number of pages for the *individual* by the number of pages for *both names*, as evidence of a *directed tie* if it reaches a threshold.

- **Associating Researchers with Topics:** For the Semantic Web community study, the system associates scientists with manually collected research topics from ISWC conference proceedings. The strength of association is calculated as the number of pages where the person's name and the topic co-occur, divided by the total pages about the person. Expertise is assigned if this value is at least one standard deviation higher than the mean for that concept. This builds on work by Mutschke and Quan Haase who clustered keywords from bibliographic records [MH01].
- **Person Name Disambiguation:** This remains the biggest technical challenge in social network mining.
    - **Problems:** Person names exhibit polysemy (common names like "Martin Frank" or "Li Ding" returning results for multiple people) and synonymy (different variations or international characters for the same person, like "Jim Hendler" vs. "James Hendler"). Web coverage can also be skewed (e.g., "George Bush the president" overrepresented compared to "George Bush the beer brewer"). This is a web-scale problem, common on the Web but rare in corporate settings.
    - **Approaches:**
        - **Bekkerman and McCallum [BM05]:** Use limited background knowledge (a list of related names) and cluster combined search results based on hyperlinks, common links, or content similarity to disambiguate appearances.
        - **Bollegala, Matsuo, and Ishizuka [BMI06]:** Apply content similarity clustering and then mine key phrases from the resulting clusters. These key phrases can be added to search queries to reduce results to the target individual (e.g., adding "beer" for "George Bush the beer brewer").
        - **Author's Work:** For the Semantic Web community, a fixed disambiguation term (e.g., "Semantic Web OR ontology") is added to queries. This is a safe limitation, focusing on the Semantic Web context, and balances precision (which Bollegala's method might improve) with recall (important for rare co-occurrences).
- **Average Precision Method:**
    - **Sophisticated Approach:** Considers the order in which search engine returns documents for a person. It assumes names of other persons appearing higher in the search results represent more important contacts.
    - **Calculation:** For a directed link from person 1 to person 2, it considers an ordered list of pages for person 1 and a set of pages for person 2 (the relevant set). It defines `rel(n)` as 1 if the

    document at position `n` is in the relevant set, and 0 otherwise. Precision at position `n` (`P(n)`) is the sum of relevant documents up to `n` divided by `n`. Average precision (`Pave`) is the average of precision at all relevant positions.
  - **Scalability:** More scalable, requiring only one download of top-ranking pages per author (linear in queries vs. quadratic for simple co-occurrence).
  - **Drawback:** Most search engines limit results (e.g., to a thousand pages), potentially missing pages for contacts of highly popular individuals.
- **Limitations and Assumptions of Statistical Methods:** A single link or co-occurrence might not be strong evidence, and not all co-occurrences are meaningful (e.g., corporate phone book entries or citation lists). These methods are statistical, assuming that the effects of uneven weights and spurious occurrences are mitigated by large numbers.

Discussion on e-Social Science e-social science represents a movement in social sciences towards reusing electronic data for social studies.

- **Driver:** The widespread digitalization of modern life, including the shift of much communication and social interaction online.
- **Opportunities:** Electronic records of online social activity offer social scientists tremendous opportunities to observe communicative and social patterns at much larger scales than previously possible, aided by advances in computer technology for handling large data volumes.
- **Automation Benefits:** Removing the human element from data collection allows for sampling data at arbitrary frequencies and observing dynamics more reliably. Traditional longitudinal studies, often limited to two time points due to prohibitive human effort, can be expanded.
- **Diversity of Settings:** The expanding online universe allows for studying a growing variety of social settings. Different data sources have unique extraction methods and limitations in their use as proxies for social phenomena.
- **Web Content Extraction (Author's Focus):** Information Retrieval-based methods for extracting social networks from Web page content offer generality, usable for various social units (individuals, institutes) and specific settings. The only requirement is significant web coverage of the network to produce reliable results despite noise (e.g., spurious co-occurrences).
- **Key Technical Hurdle:** Person name disambiguation, which involves matching references to actual persons, is the most important technical challenge. Names are neither unique nor single, leading to problems of polysemy and synonymy.

- **Semantic Web Solution:** The Semantic Web is designed to solve this problem. Websites like LiveJournal that export data in semantic formats don't suffer from this, as machine-processable descriptions with unique identifiers are attached to web pages. This technology can also be applied to merging multiple electronic data sources.

Knowledge Representation on the Semantic Web

Introduction The core idea of the Semantic Web is to represent information using formal languages that computers can process and reason with. By capturing existing Web information and adding metadata (additional descriptions of Web resources), machines can perform intelligent tasks like combining information from multiple sources for analysis.

- **Roots in AI:** Knowledge representation and reasoning are central to the Semantic Web, drawing heavily from decades of Artificial Intelligence research, particularly in expert systems. Concepts like separating domain knowledge from task knowledge, which enhances reusability, emerged from this field [SAA+99, Ste95]. For instance, a diagnostic system could be reused across different domains by simply swapping its underlying knowledge base.
- **Knowledge Representation Languages:** Expert systems developed logic-based languages like KIF and OCML for describing both domain and task knowledge. Standardization of these languages facilitated knowledge exchange and reduced development effort by increasing reusability.
- **Ontologies (AI Definition):** Domain models that captured the agreement of a group of experts and were formalized for reusability came to be known as "ontologies". (Note: The term "ontology" originated in philosophy, studying being or existence).
- **Comparison to Software Engineering:** Formal modeling is crucial in software engineering (e.g., UML, E/R models for database design) for requirement specification and design. These serve to gain agreement with domain experts and input for software/database design. Standardized languages ensure unambiguous interpretation and interoperability.
- **Semantic Web Context:** While modeling goals are similar, the Semantic Web's context of connecting knowledge bases on a web-wide scale necessitated the design of new languages. RDF and OWL are explained by comparing them to familiar techniques like E/R, relational models, UML class models, and XML.

- **Learning Curve:** Software engineers and web developers often find RDF and OWL challenging due to their grounding in formal logic, which is not widely covered in Computer Science curricula. However, they can be understood through familiar concepts of software and database design, as all these models are based on simple statements about objects, properties, and values. A basic understanding of logic is primarily needed for the concept of semantics.
- **Authoritative Resources:** For in-depth information on RDF and OWL, W3C specifications and documents like the RDF Primer and OWL Guide are recommended [MM04, SWM04]. Academic books and practical programming guides are also available [AvH04, Pow03].

4.1 Ontologies and their Role in the Semantic Web It's important to differentiate ontologies from their specific role in the Semantic Web architecture, as ontologies existed long before the Semantic Web.

- **4.1.1 Ontology-based Knowledge Representation:**
  - **Semantic Web Goal:** To extend unstructured information with machine-processable descriptions of meaning (semantics) and provide missing background knowledge.
  - **Key Challenge:** Ensuring a shared interpretation of information. Related information sources should use consistent concepts or have a mechanism to determine equivalences. Ontologies and ontology languages are the enabling technology.
  - **Definition of an Ontology (AI Context):** An ontology is defined as "a shared, formal conceptualization of a domain," detailing concepts and their relationships [Gru93, BAT97].
  - **Two Special Characteristics leading to Shared Meaning:**
    1. **Formal Languages with Well-Defined Semantics:** Ontologies must be modeled using languages with formal semantics, such as RDF [MM04] and OWL [MvH04], which have standardized syntaxes and logic-based formal semantics. When practitioners use the term "ontology" in the Semantic Web context, they often refer to models described in these languages.
    2. **Shared Understanding within a Community:** Ontologies are built upon an agreement among community members regarding the concepts, relationships, and their usage within a domain. This implies that there is no such thing as a "personal ontology". A database schema or UML class diagram, while a conceptual model, is not an ontology unless it represents a commitment shared by a broader community.

- **Complexity Spectrum of Ontologies:** The definition doesn't prescribe the level of detail or minimal expressivity required. Ontologies vary greatly in complexity.
    - **Simplest Structures:**
        - **Glossaries/Controlled Vocabularies:** An agreement on the meaning of a set of terms, facilitating uniform descriptions (e.g., for support incidents). A "mere bag of words" does not meet the criteria of shared meaning.
        - **Semantic Networks:** Graphs showing how terms are related to each other.
        - **Thesauri:** Richer structures that describe hierarchies (broader-narrower relationships, which are transitive) and related terms/aliases. These are the simplest where logic-based reasoning can be applied.
    - **Semantic Web Context:** Often assumes an ontology contains at least a concept hierarchy (subclass relationships). Weaker models like folksonomies are sometimes excluded but can have hierarchies and tag relationships extracted.
    - **Lightweight vs. Heavyweight Ontologies:**
        - **Lightweight Ontologies:** Typically distinguish between classes, instances, and properties but contain minimal descriptions. Examples include models expressed in RDF(S) and OWL Lite.
        - **Heavyweight Ontologies:** Allow for more precise descriptions, defining how classes are composed and providing richer constructs to constrain property application. They use the full expressivity of First Order Logic (FOL) to detail instances and relationships. Examples include OWL DL, OWL Full, and rule-based systems.
        - **Benefits of Increased Detail:** More constrained descriptions reduce misinterpretation by human readers. Closer to the "heavyweight" end, computers play a larger role in reasoning tasks like checking consistency of term usage, classifying concepts (finding subclass relations), and checking for equivalent terms.
    - **Expressivity and Computational Complexity:** More expressive languages can capture simpler models. However, a language that is too expressive (offers unused features) can hinder modeling. Each language offers features tailored to specific tasks or domains. Less expressive languages

generally have lower guaranteed computational complexity for reasoning.

        - **Practicality on the Web:** Most common Web ontologies are lightweight due to the need to serve many applications with diverse goals. Widely shared Web ontologies tend to be small, containing only broadly agreed-upon terms. Large, heavyweight ontologies are found in specialized expert systems for focused domains (e.g., life sciences, engineering) with formalized processes and vocabularies. There is an unavoidable trade-off between formality and sharing scope: widely shared models tend to be shallow, while limited-scope models can be more detailed [MA04].

- **4.1.2 Ontologies and Ontology Languages for the Semantic Web:**
    - **Special Role in Architecture:** Ontologies are fundamental to the Semantic Web's architecture, which motivates the design of RDF and OWL for distributed and open environments.
    - **Realization of the Semantic Web:** Achieved by annotating existing Web resources with ontology-based metadata and by exposing database content using standard ontology languages.
    - **Globally Unique Identifiers (URIs):** Semantic Web ontology languages identify concepts using URIs. These URIs allow data sources to point to concepts or relationships in external, public ontologies. This creates a "web" of knowledge sources, similar to hyperlinks among HTML pages. Since URLs are also URIs, ontologies can describe existing Web resources.
    - **Data Publication:** Metadata and schemas can be published as RDF or OWL documents online (possibly generated dynamically from databases or web services) or by providing access via the standard Semantic Web query language and protocol, SPARQL. Schema and instance data can be published separately or mixed.
    - **Semantic Web Applications:** These applications collect or query data sources, then aggregate and reason with the results. If an application knows the schema (implying commitment to an ontology and agreed meaning), it can directly consume the information. A key task is then aggregating instance data and finding equivalent instances across sources, a process where formal semantics is crucial.
    - **Ontology Mapping:** When creating or reusing ontologies, one might create from scratch or extend existing ones. However, uncoordinated development can lead to different communities creating overlapping ontologies (e.g., multiple ontologies describing scientific publications). In such cases, determining how classes in different ontologies relate (e.g., "Article" in one is

"JournalPublication" in another) is the task of *ontology mapping*, a complex problem not covered in the source.
    - **Importance of Machine-Processable Semantics:** Formal and detailed ontologies are vital for accurately capturing the intended meaning of concepts and relationships, especially in the wide scope of the Web. Precise definitions prevent misinterpretation by humans and, crucially, enable machines to "understand" terms (e.g., to automatically map equivalent concepts). Automating such mapping is difficult but indispensable at Web scale.

4.2 Ontology Languages for the Semantic Web The World Wide Web Consortium (W3C) has standardized RDF and OWL as ontology languages.

4.2.1 The Resource Description Framework (RDF) and RDF Schema

- **Purpose:** RDF was initially created to describe resources on the World Wide Web but is domain-independent, capable of modeling both real-world objects and information resources. It is a primitive modeling language, forming the foundation for more complex languages like OWL.
- **Primitives:**
    - **Resources:** Everything that can be potentially identified and described is modeled as a resource. Resources are identified by URIs (Uniform Resource Identifiers) or are "blank". URIs have a special syntax [BLFM98], with `http:` URIs being common identifiers even for non-Web resources. Blank resources (blank nodes) act as existential quantifiers, having an identity but an unknown or irrelevant identifier.
    - **Literals:** Character sequences that can have optional language and datatype identifiers.
- **Statements (Triples):** Expressions in RDF are formed as "triples" of the form (subject, predicate, object).
    - The subject must be a resource (URI-identified or blank).
    - The predicate must be a URI.
    - The object can be either a resource or a literal. Literals are only allowed as objects of statements.
- **Graph Model:** RDF statements inherently form a directed, labeled graph. Nodes represent subjects and objects (labeled with URIs, literals, or left blank), and edges represent predicates (labeled with the property's URI). The object of one statement can become the subject of another, creating interconnected graphs.
- **Syntax:**

    - **Turtle Syntax:** Figure 4.3 illustrates RDF statements in Turtle syntax, one of several notations. It's preferred for human readability.
    - **RDF/XML Syntax:** The first and most widely used syntax, it's XML-based and can be processed by XML tools.
    - **Namespaces:** Both Turtle and RDF/XML allow abbreviating URIs using namespaces, which are a syntactic convenience, not an RDF feature.
- **RDF Vocabulary:** (See Table 4.1)
    - **Basic Constructs:** `rdf:type` (assigns a type to a resource), `rdf:Property`, `rdf:XMLLiteral`.
    - **Collections:** `rdf:List`, `rdf:Seq`, `rdf:Bag`, `rdf:Alt` (for different types of collections), `rdf:first`, `rdf:rest`, `rdf:nil` (for linked lists), `rdf:n` (for enumerating elements), `rdf:value`. The `rdf:n` properties theoretically make the vocabulary infinitely large, but this isn't a practical issue.
    - **Reification:** `rdf:Statement`, `rdf:subject`, `rdf:predicate`, `rdf:object` (for making statements about statements).
- **RDF Schema (RDFS):** RDF itself has minimal descriptive power and is almost always used with RDF Schema. RDFS is a simple extension that defines a modeling vocabulary including notions of classes and subclasses. It allows connecting classes and properties by specifying the `rdfs:domain` and `rdfs:range` of properties.
    - **RDFS Vocabulary:** (See Table 4.2) Includes `rdfs:domain`, `rdfs:range`, `rdfs:Resource`, `rdfs:Literal`, `rdfs:Datatype`, `rdfs:Class`, `rdfs:subClassOf`, `rdfs:subPropertyOf`, and documentation terms like `rdfs:comment`, `rdfs:label`.
    - **Unnatural Division:** The division of terms between RDF and RDFS namespaces is "very unnatural" (e.g., `rdf:type` in RDF namespace without classes there, `rdfs:Literal` in RDFS namespace though literals are an RDF feature). For practical purposes, "RDF ontology" or "RDF data" usually refers to an RDF Schema ontology.
- **Flexibility of RDF:**
    - **Unified Modeling:** Uses the same subject/predicate/object model to describe both instances (data) and classes (schema). While often stored separately, they form a single graph when combined.
    - **No Clear Instance/Class Separation:** It's not always easy to separate instance and class descriptions, and they can be stored together. Conceptual ambiguities exist (e.g., a "Weekday" being a class of five days, blurring instance and class data). This ambiguity is reflected in the term "ontology" sometimes referring to classes only, sometimes to instances and classes together.

- **Self-Referential Language:** RDF's language constructs (e.g., `rdfs:Class`) are treated like any other user-defined term. This means statements can be made *about* the language elements themselves (e.g., declaring a domain-specific labeling property like `foaf:name` as a subproperty of the more general `rdfs:label`).
- **Metamodeling:** RDF does not enforce a strict separation between classes, instances, and properties, allowing for metamodeling (creating classes of instances), which is useful for practical knowledge (e.g., "species" as a class of animal classes). `rdf:type` can be used on instances and classes, and `rdfs:label` on instances, classes, and properties. Classes of properties can also be used.
- **Loose Interpretation of Constraints:** RDF language constructs are not interpreted as strictly as one might expect. For example, even if `foaf:knows` has a range of `foaf:Person`, stating that `example:Rembrandt foaf:knows mailto:saskia@example.org` does not cause a logical contradiction; it's interpreted as a hint that `mailto:saskia@example.org` is *both* an email address *and* a `Person`. This highlights the importance of understanding RDF(S) semantics precisely.

- **RDF and the Notion of Semantics:**
  - **Need for Reliable Meaning:** For RDF(S) to convey knowledge across the Web, its meaning (and thus the meaning of ontologies) must be defined reliably. This is critical for machine reasoning, such as checking if a statement follows from existing ones or if contradictions exist.
  - **Model-Theoretic Semantics:** The meaning of RDF(S) constructs is anchored in model-theoretic semantics, a common approach [Hay04]. This involves establishing a mapping from the ontology's symbols to a meta-model (an independently existing reality) where the truth of propositions is uniquely determined. This mapping is called an "interpretation".
  - **Constraints on Interpretations:** RDF constructs impose constraints on possible interpretations, excluding unintended ones. For example, `owl:differentFrom` can be used to exclude an interpretation where symbols for two distinct persons map to the same object.
  - **Axiomatic Triples and Inference Rules:** The model theory of RDF(S) includes "axiomatic triples" (always true in any RDF(S) interpretation, e.g., `rdf:Property rdf:type rdfs:Class`) and defines constraints as a set of rules. For instance, `rdfs:subPropertyOf` implies transitivity and type assignments for the properties involved.

- **Open World Assumption (OWA):** A key feature of RDF semantics is the open world assumption. This means that based on a single document, one cannot assume a complete description of the world or even the explicitly described resources.
- **Monotonicity:** Adding new knowledge to an RDF knowledge base cannot invalidate previous inferences. It is "altogether impossible to create logical inconsistencies in RDF" except for a datatype clash. Consequently, "RDF validation" in the sense of contradicting previous statements makes no sense, as RDF Schema statements merely add additional information. The semantics are specified as inference rules; for example, if `foaf:knows` has range `Person` and Rembrandt `foaf:knows` Pluto the dog, it's inferred that Pluto is *also* a Person.

- **SPARQL: Querying RDF Sources Across the Web:**
  - **Standardization:** While several ontology databases (triple stores) and their query languages existed, the W3C developed SPARQL as a standard query language and protocol for interacting with RDF sources.
  - **Functionality:** SPARQL allows selecting and retrieving parts of an RDF graph by matching a graph pattern specified in the query and returning variable values.
  - **Interoperability:** The SPARQL specifications are expected to significantly enhance the interoperability of Semantic Web applications.
  - **Broader Access:** Non-RDF databases can also expose their content via a SPARQL interface, opening them to the Semantic Web world, and this "wrapping" process can be partially automated.

4.2.2 The Web Ontology Language (OWL)

- **Purpose:** OWL was designed to incorporate constructs from Description Logics (DL) into RDF, significantly enhancing RDF Schema's expressiveness for characterizing classes and properties. DLs have formal semantics mapped to First Order Logic (FOL) and have been studied for their expressivity-reasoning efficiency tradeoffs. OWL was designed to map to a DL with tractable (efficient) reasoning algorithms.
- **Three Languages:** OWL is a family of three languages with increasing expressiveness: `OWL Lite`, `OWL DL`, and `OWL Full`. They are extensions of each other both syntactically and semantically (e.g., an OWL Lite document is a valid OWL DL document with the same semantics). Their vocabularies extend each other, and more expressive languages relax usage constraints.

- **Relationship to RDF(S):** While commonly believed to be a simple extension of RDF(S), only `OWL Full` truly includes RDF(S) (`RDF(S) ⊆ OWL Full`).
- **OWL DL:**
    - **Design Target:** The original target of standardization, directly mapping to an expressive Description Logic.
    - **Reasoning:** OWL DL documents can be directly used by DL reasoners for inference and consistency checking.
    - **Semantics:** Constructs are familiar, but semantics can be surprising due to the open world assumption [RDH+04].
    - **Limitations:** DLs do not permit much of the representation flexibility found in RDF (e.g., treating classes as instances or defining classes of properties). Not all RDF documents are valid OWL DL documents, and even OWL terms have limited usage. For example, in OWL DL, extending constructs within the RDF, RDF Schema, and OWL namespaces is not allowed. OWL introduces a distinct `owl:Class` concept (as a subclass of `rdfs:Class`) to clarify its more limited notion of a class. It also introduces disjoint classes for `owl:ObjectProperty` (takes resources as values) and `owl:DatatypeProperty` (takes literals as values).
- **OWL Full:**
    - **"Limitless" OWL DL:** Any RDF ontology is a valid `OWL Full` ontology and retains its semantics when interpreted as `OWL Full`.
    - **Decidability:** `OWL Full` is undecidable, meaning reasoners might run infinitely in the worst case.
- **OWL Lite:**
    - **Lightweight Sub-language:** A less expressive but more efficient DL language than `OWL DL`.
    - **Limitations:** Shares the same limitations on RDF usage as `OWL DL` and lacks some `OWL DL` terms.
- **Summary of Relationship to RDF:** RDF documents are not necessarily valid `OWL Lite` or `OWL DL` ontologies, despite common perception. Converting a typical RDF or `OWL Full` ontology to `OWL DL` is a "tedious engineering task" involving steps like declaring properties as object or datatype properties and importing external ontologies (mandatory in OWL, not RDF). It often requires fundamental modeling decisions.
- **Usage:** Most existing web ontologies make limited use of OWL due to simpler needs and OWL's inability to express general rule-based knowledge. However, OWL's additional expressivity is necessary for modeling complex domains like medicine or engineering, particularly for classification tasks (determining a class's position in a hierarchy based on its description).

- **Query Language:** There is currently no standard query language for OWL, though a proposal exists [FHH04].

4.2.3 Comparison to the Unified Modeling Language (UML)

- **UML Context:** Primarily used in the requirements specification and design phases of object-oriented software for enterprise applications [Fow03].
- **Chief Difference (Modeling Scope):** UML has specific modeling primitives for objects in information systems, covering static attributes, associations, and dynamic behavior (e.g., interfaces, functions). RDF(S)/OWL are more general.
- **Overlap:** Despite differences in scope and primitives, there's significant overlap in expressiveness between object-oriented domain models and ontological models. Properties taking primitive values are modeled as datatypes, others as associations in UML (Figure 4.8 illustrates this). UML is mainly a schema definition language, with limited instance modeling.
- **Unique Features of RDF/OWL:**
  - **Less Constrained:** RDF modeling is generally less constrained than UML, meaning many RDF models have no UML equivalent.
  - **More Primitives:** OWL DL offers more primitives than UML, such as disjointness, union, intersection, and equivalence of classes.
  - **Defined Classes:** OWL allows defining classes with necessary and sufficient conditions for instance membership.
  - **First-Class Properties:** RDF/OWL treat properties as first-class citizens: they are global, not belonging to any single class, and can be used with multiple classes. Properties can also be supplied as values of other properties.
  - **Subproperties:** Defining subproperties is straightforward in RDF/OWL, less so in UML.
  - **Metamodeling:** Classes can be treated as instances.
  - **Reification:** RDF reification is more flexible than UML's association class mechanism, as it can reify statements involving literal values (which are modeled as attributes in UML, and association classes cannot attach to attributes).
  - **URIs for Identification:** All non-blank RDF resources are identified with URIs; UML classes, instances, and attributes do not inherently have such IDs.
  - **Multiple Types:** Instances can, and often do, have multiple types (distinct from multiple inheritance, supported by both).
- **Unique Features of UML:**

    - **Relationship Roles:** UML has the notion of relationship roles, which is absent in RDF/OWL.
    - **N-ary Relations:** UML allows n-ary relations, which are not native to RDF but can be re-represented.
    - **Part-Whole Relations:** Two common types (aggregation and composition) are available in UML, which can be partially re-modeled in OWL.
    - **Attributes vs. Associations Distinction:** UML distinguishes attributes from associations differently than OWL's datatype vs. object properties. UML attributes can have instances as values, while OWL datatype properties only take literal values. However, OWL applies cardinality constraints to both datatype and object properties, and reification applies to statements with datatype properties.
- **Semantics:** A direct comparison of UML class model semantics and RDF/OWL model semantics is difficult because UML's semantics are given by its meta-language (MOF - Meta Object Facility), and there is no direct mapping to RDF/OWL. A key difference is that MOF is built on a *closed world assumption*, while RDF/OWL assume an *open world*.
- **Interoperability:** Given that enterprise software often follows OO methodologies, considering the relationship between UML and OWL is important. It's natural to generate object models from ontological models or vice-versa. Code generators exist for creating object models in languages like Java from RDF(S) and OWL. Tools also facilitate conversion from UML class models to OWL. The Ontology Definition Metamodel (ODM) is a UML profile for OWL ontologies, acting as a visual modeling language for OWL based on UML.

4.2.4 Comparison to the Entity/Relationship (E/R) Model and the Relational Model

- **E/R Model Context:** Commonly used for information modeling at the data storage layer of applications, as it maps easily to the relational model used in database management systems (RDBMS). It's simpler than UML.
- **E/R Constructs:** Focuses on modeling information based on relationships, characterized by arity, participating entity sets, cardinalities, and dependency. Attributes can be attached to relationships, and relationships can be included in other relationships, similar to reification in RDF and UML.
- **Entity Sets:** Roughly correspond to classes in UML. The E/R model is less concerned with modeling entity sets, with "generalization" being the only predefined relationship type between them, whose semantics are limited to attribute inheritance (lower-level entity sets inheriting attributes

from higher levels). Unlike UML and RDF, generalization between relationships (`rdfs:subPropertyOf`) is not allowed.

- **Keys:** A special feature of E/R is the notion of keys (sets of attributes uniquely identifying an entity) and weak entities (identified by attributes plus relations to other entities). Single-attribute keys can be modeled in RDF using functional and inverse functional properties, but complex keys cannot be accurately modeled in RDF/OWL.
- **Instance Representation:** E/R diagrams represent only the schema, not instances. Relational tables can face problems identifying instances if primary keys are missing (e.g., Saskia without an `mbox` primary key). RDF naturally solves this with inherent identifiers (URIs or bNode IDs), allowing unique instance identification even with partial descriptions.
- **Mapping to Relational Databases:** Although conceptually distant, the relationship between E/R models and ontologies is important for exposing RDBMS data via a web interface as RDF. Establishing a close mapping between a relational database schema and an ontology is sensible. However, automating this mapping is difficult because E/R diagrams can be mapped in multiple ways to the simpler relational model, which is often "tweaked" for performance. Consequently, relational models extracted from existing databases usually require significant post-processing and enrichment to form an accurate domain model [VOSS03].

4.2.5 Comparison to the Extensible Markup Language (XML) and XML Schema

- **Purpose:** XML is widely used for structured information exchange, making its purpose most similar to RDF.
- **Data Models:**
    - **RDF:** A directed graph.
    - **XML:** A directed, ordered tree, originating from SGML (Standard Generalized Markup Language) for marking up text documents where nesting and ordering matter.
    - **Comparison:** Every tree is a graph, but representing XML models in RDF is not trivial, largely due to XML's strict ordering and hierarchical nature. RDF's more relaxed graph model is better suited for conceptual domain models with rich cross-taxonomical relationships that are difficult to force into a single, unique tree structure (e.g., social networks).
- **Schema Languages:**
    - **XML Schema/Relax NG:** Define element types and attributes, but critically, also *prescribe syntax* (how elements can be nested). XML documents can be validated against a schema at a purely syntactic level.

    - **RDF Schema/OWL:** Do not impose direct constraints on the graph model but affect the possible interpretations of metadata. Consistency is checked by searching for possible interpretations (inconsistency if none exist). RDF and OWL provide a more refined set of constructs (classes, instances, properties) compared to XML's vaguer notions (elements, attributes, nesting).
- **Semantics vs. Syntax:**
    - **XML Focus:** XML requires agreement on *both syntax and semantics*. Different XML representations can convey the same meaning but have different syntax, making applications incompatible if they rely on specific parsing.
    - **RDF Focus:** RDF has a clear advantage as agreement only needs to concern the representation of individual statements (the simple subject/predicate/object model), making it more flexible for web-based data interchange.
- **Tooling and Legacy:** XML boasts a rich ecosystem of tools (XML databases, XQuery, XSLT, editors, parsers). The W3C adopted XML as a notation for RDF and OWL due to this legacy.
- **The Semantic Web Layer Cake (Figure 4.11):** This diagram shows the Semantic Web as layers built upon existing standards like XML, URIs, and Unicode. However, the author argues it "obfuscates the true relationship between XML and ontology languages".
    - **RDF builds on XML?** This is only true in that RDF models can be exchanged using the `RDF/XML` format.
    - **Serialization Sensitivity:** `RDF/XML` allows multiple serializations for the same RDF graph (depending on abbreviations, statement order), making XML queries and XSL transformations sensitive to a particular serialization. XML editors are only helpful for checking well-formedness of `RDF/XML` files.
- **XML to RDF Conversion:**
    - **Difficulty:** Re-representing XML schemas and data using RDF is more challenging than the layer cake suggests.
    - **"Striped Syntax":** RDF/XML mandates a "striped syntax" where instances and properties are nested correctly. Many XML schemas don't conform to this (e.g., an XML document lacking a level of elements between `shipTo` and `name` where `name` is a property of an omitted `Person` class).
    - **Automation:** Full automation of XML to RDF conversion is not possible in the general case. XSLT transformations can be designed for specific schemas.
    - **GRDDL:** The GRDDL (Gleaning Resource Descriptions from Dialects of Languages) specification provides a standard way to attach such transformations to XML/XHTML documents, though

the agreement on metadata representation and extraction is left to authors.
  - **RDF/A:** Another W3C proposal, RDF/A, offers a generic, domain-independent way to embed RDF metadata in XHTML using attributes and special elements [AB06].
  - **Designing New Formats:** For new formats, a DTD or XML Schema can be created to conform with RDF/XML requirements (e.g., RSS 1.0, which can be processed by both XML and RDF tools).
  - **XML Schema to RDF/OWL Conversion:** Not trivial because XML and XML Schema have notions (e.g., distinction between attributes and terminal elements, natural ordering of elements, uniqueness, keys, integrity constraints) that RDF/OWL lack or cannot trivially express.
- **Summary:** RDF/OWL are more flexible, expressive, and conceptually cleaner than XML. Their introduction represents a "conscious break" from the XML world by the W3C. However, XML's enormous deployment advantage means these paradigms will coexist for years.

4.3 Discussion: Web-based Knowledge Representation RDF and OWL have been specifically engineered for knowledge representation in distributed settings like the Web.

- **Distribution:** RDF/OWL allows distributing data and schema information across the Web. Like XML, RDF documents can reside anywhere and reference other vocabularies using globally unique URIs. URIs with `http:` protocol can be looked up to provide authoritative descriptions. RDF Schema and OWL vocabularies include terms (`rdfs:seeAlso`, `rdfs:definedBy`, `owl:import`) that link RDF documents together, forming a "Web of RDF documents" akin to HTML documents linked by hyperlinks.
- **Formal Semantics:** The existence of formal semantics ensures unambiguous interpretation of the language, which is critical for sharing information across the Web. Meaning is defined by constraints placed on possible mappings to an interpretation domain (model-theoretic semantics), informally thought of as an independently existing reality. This allows the language to exclude unwanted interpretations.
- **OWL's Non-Monotonicity and Inconsistencies:** A clear difference from RDF is OWL's non-monotonic logic and the possibility of creating inconsistencies. Because negation is part of the language, new knowledge can contradict previous assertions. OWL functions more like a constraint language, where logical inconsistencies (e.g., unsatisfiable classes in the schema, or instance violations of cardinality constraints) can be detected.

OWL reasoners help localize these sources of inconsistency, though automating their removal is still a research problem.

In summary, RDF/OWL are a set of languages designed for information sharing across the Web, varying in expressivity. While they differ in purpose from other well-known knowledge representation formalisms in Computer Science, they share many primitives with E/R, UML, and XML models. The ability to accurately represent and share the meaning of information through formal semantics is crucial for information sharing in heterogeneous settings where other mechanisms for shared understanding are unreliable. A semantic-based representation is also a vital first step towards data integration when data is distributed and under diverse ownership and control.

I've provided a comprehensive overview of the material you asked for. Would you like me to test your comprehension of this material through a short quiz or a scenario, or perhaps explore some other related material from the sources? For instance, we could delve deeper into the specific features of RDF or OWL, or discuss the challenges of information extraction in more detail.